MLP Task 2

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Testing Results

Test Number	Activation Function	Hidden Nodes	Layers	Learning Rate	Epochs	Bias	Train Accuracy	Test Accuracy
Test 1	Sigmoid	3,4	2	0.01	1000	Yes	98.89%	100.0%
Test 2	Tanh	5	1	0.001	5000	Yes	100.0%	98.33%
Test 3	Sigmoid	4	1	0.1	500	No	100.0%	98.33%
Test 4	Tanh	4,4	2	0.05	1000	Yes	100.0%	100.0%
Test 5	Sigmoid	2	1	0.001	200	Yes	65.56%	66.67%

Test 1 (Sigmoid, 2 hidden layers: 3, 4):

Conclusion: The Sigmoid activation successfully classified the test set with 100% accuracy, demonstrating a classic, gradual convergence curve over 1000 epochs due to its smaller gradient updates.

Test 2 (Tanh, 1 hidden layer: 5):

Conclusion: Using Tanh on Z-score scaled data resulted in instant memorization of the training set (100% accuracy) and highly stable generalization (98.33% test accuracy) across 5000 epochs.

Test 3 (Fast Sigmoid, No Bias):

Conclusion: By removing the bias and increasing the learning rate to 0.1, the network converged much faster (500 epochs), proving that the 5 selected features are distinct enough to be linearly separated without a bias shift.

Test 4 (Deep Tanh, 2 hidden layers: 4, 4):

Conclusion: Tanh handled the deeper two-layer architecture flawlessly, avoiding the vanishing gradient problem entirely to achieve a perfect 100% accuracy on both training and unseen test data.

Test 5 (Underfitting Example):

Conclusion: Restricting the architecture to only 2 hidden nodes alongside a tiny learning rate (0.001) and low epoch count (200) resulted in severe underfitting (66.67% accuracy), as the network lacked the internal capacity and time to learn the complex decision boundaries

Visualizations

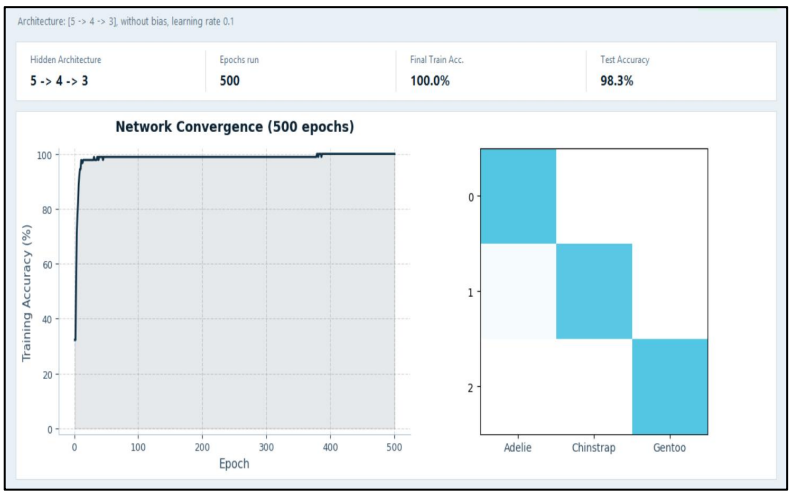
Test 1



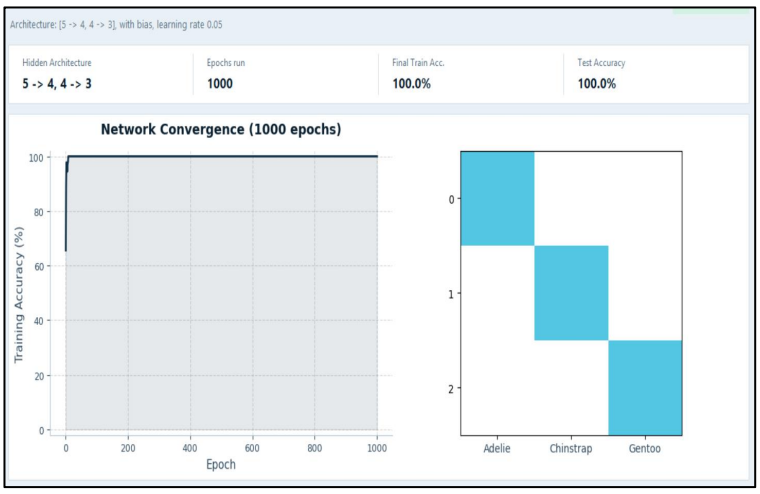
Test 2



Test 3



Test 4



Test 5

